Artificial Intelligence Lab

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LAB 05

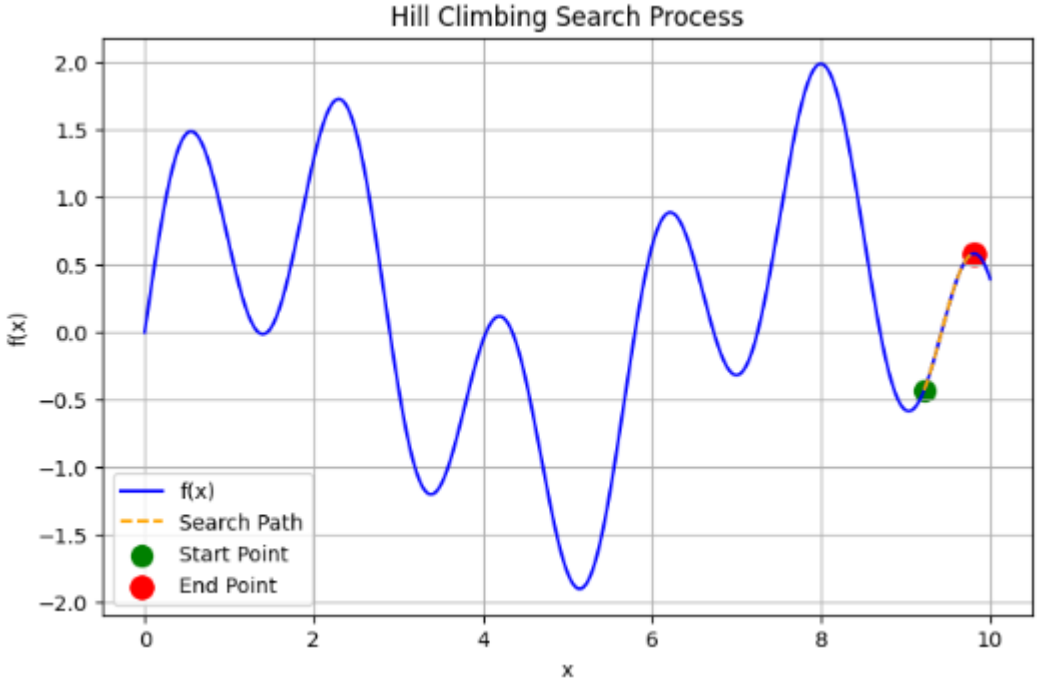
1. Consider the following objective function (landscape) defined over the domain $x \in [0, 10]$:

$$f(x) = \sin(x) + \sin\left(\frac{10x}{3}\right)$$

   Your task is to find the global maximum of this function. You are required to trace the execution of two different local search algorithms.

   1. Hill Climbing Search
   2. Simulated Annealing Search

      Visualize the graph and searching process in both of the cases.



2. Solve the 8-Queen problem using the local search algorithms

   1. **First-choice** Hill Climbing Search
   2. Simulated Annealing Search

      Design the required objective function as per your choice.